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An Improved Electrical Equivalent Circuit Model for Dynamic Moving Coil Transducers

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ABSTRACT

A series combination of inductor and resistor is traditionally used to model the blocked electrical impedance of a dynamic moving coil transducer, such as a loudspeaker driver. In practice, semi-inductive behaviour due to eddy currents and 'skin effect' in the pole structure as well as transformer coupling between the voice coil and pole piece can be observed, but are not well represented by this simple model. An improved model using only a few additional elements is introduced to overcome these limitations. This improved model is easily incorporated into existing equivalent circuit models. The development of the model is explained and its use is demonstrated. Examples yielding more accurate box response simulations are also shown.

1. INTRODUCTION

Low frequency lumped parameter loudspeaker models generally make use of 'equivalent circuits' in which quantities such as mass and velocity take on analogous electrical parameters such as inductance and voltage. Using the 'admittance analogy', the electrical side of the

system is 'coupled' to the mechanical side by an ideal transformer with a turns ratio of $Bl:1$, where Bl is the force factor of the speaker's motor. The input to the system is voltage and the output of the system is cone velocity u , as seen in Fig. 1. Traditionally, when performing these calculations, the radiation resistance of the cone operating into air is ignored since it has little influence on the motion of the diaphragm. The air load

is regarded as part of the equivalent mass of the cone (M_{MS}). R_E and L_E are the resistance and the inductance of the voice coil, C_{MS} is the compliance of the total suspension and R_{MS} represents the mechanical resistance in the system [1, 2]. An improved model of the electrical impedance of the loudspeaker will be developed and its impact on enclosure simulations will be explored. The focus is on improvements to the electrical side of the equivalent circuit, both to make the model agree more closely with measurement, and to have the elements in the model represent understood behaviour (physical modelling).

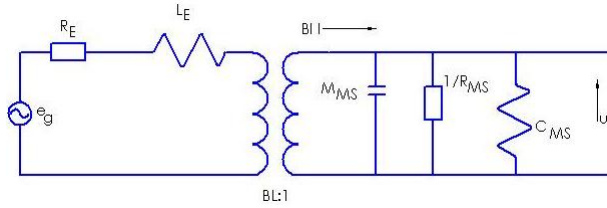


Fig. 1 Equivalent circuit diagram for a loudspeaker using the 'admittance' analogy.

2. IMPEDANCE

2.1. Blocked and Motional Impedance

The electrical impedance of a loudspeaker, $Z_s(f)$ can be considered to consist of two parts, shown in the dashed boxes in Fig. 2.

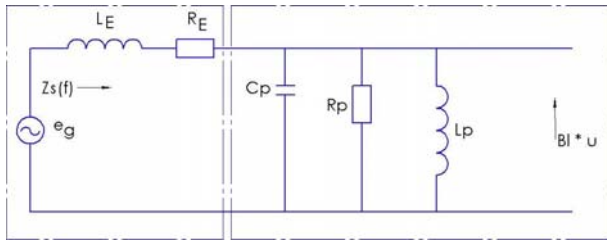


Fig. 2 Equivalent circuit diagram for a loudspeaker converted to the electrical side.

The components in the first box constitute the 'blocked' impedance, which for small signals are independent of the motion of the speaker diaphragm. The components shown in the second box constitute the motional impedance. Motional impedance occurs when the voice coil moves through the magnetic field of the motor. This

in turn, sets up an electromotive force (i.e., e.m.f., or voltage) equal to $Bl u$, where u is velocity, that works to oppose the velocity of the voice coil.

The mechanical parameters are converted to virtual electrical components by the equations shown below:

$$C_p = \frac{M_{MS}}{(Bl)^2} \quad (1)$$

$$R_p = \frac{(Bl)^2}{R_{MS}} \quad (2)$$

$$L_p = C_{MS} \cdot (Bl)^2 \quad (3)$$

$$V = Bl \cdot u \quad (4)$$

The subscript 'P' (parallel resonance circuit component) is used to designate the virtual electrical components affecting the motional impedance.

As shown in Fig. 2, the loudspeaker can be represented by a four-pole (two-port) electrical circuit. The circuit will have a resonance frequency, f_s , and at that frequency the electrical impedance will attain a local maximum, Z_0 when $1/j\omega C_p$ and $j\omega L_p$ cancel. However, in practice, very small residual reactive impedance due to L_E will move the measured resonance peak slightly upwards in frequency. Since at resonance the motional component of the electrical impedance is equal to R_p it follows that:

$$R_p = Z_0 - R_E \quad (5)$$

If the circuit shown in Fig. 1 is probed with a current i , then a voltage $Bl u$ will appear at the output terminals. If the frequency of the probing signal is f_s , then u will be maximized. Since the motional impedance is equal to R_p , then:

$$Bl = \frac{I \cdot R_p}{u_{\max}} \quad (6)$$

$u_{\max}$ could also be measured directly using a laser transducer [3].

2.2. Differences Between Measured and Modeled Impedance

It is not difficult to measure Z_0 , but as will be shown later, R_E is not exactly the same as R_{DC} . It is always slightly greater – with the consequence that Q_{ES} (the electrical Q-factor) is generally underestimated if R_E is assumed to be the same as R_{DC} . Previously, this has been correctly attributed to eddy currents [3, 4]. For the same reason, Bl , η , and SPL_{REF} – the sensitivity at f_s for $Q_{TS} = 1$ (Q_{TS} – the total Q-factor for the speaker) are all overestimated when R_E is assumed to be the same as R_{DC} [1, 2]. Consequently, in this paper, we will distinguish between R_E and R_{DC} . This will be discussed in more detail.

The shape of the loudspeaker impedance function, $Z_s(f)$ is well known (see Fig. 3). Starting near R_{DC} at very low frequencies, it is followed by a peak at the fundamental resonance (of M_{MS} and C_{MS}). Above the resonance frequency, there is an anti resonance minimum at f_{min} (primarily between L_E and $M_{MS}/(Bl)^2$ acting as a large capacitor). Above this minimum, the impedance according to the model should raise proportionally with frequency.

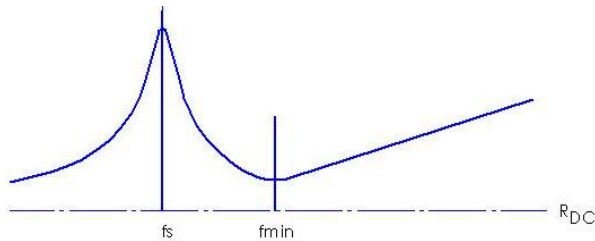


Fig. 3 Typical loudspeaker impedance. Log ordinate (dB scale).

Comparison between the impedance curve predicted by the simple equivalent circuit and the actual measured impedance reveals some differences. First, at f_{min} the impedance magnitude should be very close to R_{DC} , and the phase angle should be close to zero. In practice, however, the impedance at f_{min} is always higher than predicted. Typically for a woofer, the impedance is often more than 1 dB higher and the phase is not zero. Secondly, the measured slope of the impedance curve at

high frequencies is typically closer to 3 dB/octave rather than the 6 dB/octave we would expect if the only inductive element in the circuit were a conventional inductor. Obviously, a box simulation based on this simple equivalent circuit model will often result in significant errors.

2.3. Eddy Current Effects and Semi-Inductance

Vanderkooy [4] explains the 3 dB slope of the impedance curve as the result of eddy currents in the iron core (of the speaker's pole piece) and as the result of the 'skin effect'. As the skin depth decreases with the square root of frequency, the electrical conductivity of the iron in the pole piece is gradually reduced. Accordingly, the skin depth is

$$\delta = \sqrt{\frac{2}{\mu \sigma \omega}} \quad (7)$$

Here μ is the permeability and σ is the conductivity of the iron. If the coil and iron core were infinitely long – or in the case of a core in the form of a closed magnetic circuit – it is shown (using Maxwell's equations) that the coil will act as a "semi-inductor", even at very low frequencies. The impedance of this semi-inductor is calculated as

$$Z = K \cdot \sqrt{j\omega} \quad (8)$$

This function will rise 3dB/octave as observed. The units of K are in 'semi-Henrys' [sH]. Due to the short length of the core and the very open structure of the magnet system, L_E acts as an ordinary inductor at low frequencies, and typically becomes a semi-inductor above 100-200 Hz. The low frequency inductance of the loudspeaker voice coil is only a few times higher than its value in free air. This implies that the relative differential permeability is around 2, not around 3000 as would normally be expected for iron. This is likely the reason for the unexpectedly high transition frequency, below which the skin depth is more than $\frac{r}{\sqrt{2}}$, where r

is the radius of the pole piece. Vanderkooy did not try to make an equivalent circuit model combining these components to include the transition from inductor to semi-inductor – in a frequency range important for

precise box simulation. Developing such an equivalent circuit model is the object of this paper.

The total loudspeaker impedance Z_S as mentioned consists of two parts, the 'blocked' electrical impedance Z_E and the motional impedance Z_M . These will be treated separately below.

2.4. Modeling the 'Blocked Impedance'

Considering the geometry of the motor in a modern speaker, it is very difficult to predict the exact character of the electrical impedance of the voice coil. However, it can be measured by blocking the movement of the coil. One way to do this is by fixing the coil in place by filling the air gap with epoxy.

Simple intuition suggests that the blocked impedance – and therefore the combined inductor/semi-inductor behaviour – might be modelled by a resistor R_E in series with a parallel connection of two components: a semi-inductor K_E and an ordinary inductor L_E (see Fig. 4). It is obvious that this combination will have some of the right features. At low frequencies, it is equal to R_E . By suitable choice of the other two components, the inductor will become dominant at low frequencies and the semi-inductor will take over smoothly at higher frequencies. To be successful, it should give the correct minimum impedance at the measured f_{min} in combination with the calculated motional impedance – and generally it should duplicate the measured impedance curve.

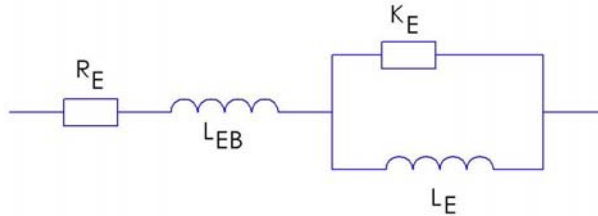


Fig. 4 Hypothetical equivalent network for the blocked electrical impedance.

The proposed equivalent network also includes an extra inductor L_{EB} , which represents the part of the coil extending above the pole piece. Therefore to a certain degree, this element is expected to behave as a normal inductor.

The blocked electrical voice coil impedance Z_E in Fig. 4 is then comprised of R_E in series with L_{EB} and with a regular inductor L_E shunted by a semi-inductor K_E . The impedance of the inductor is $j\omega L_E$. The impedance of the semi-inductor is

$$K_E \cdot \sqrt{j\omega} = K_E \frac{1+j}{\sqrt{2}} \sqrt{\omega} \quad (9)$$

in [sH]. The impedance of this combination then becomes:

$$Z_E(j\omega) = R_E + j\omega L_{EB} + j\omega L_E \parallel K_E \sqrt{j\omega} \quad (10)$$

or

$$Z_E(j\omega) = R_E + j\omega L_{EB} + \frac{j\omega L_E \cdot K_E \frac{1+j}{\sqrt{2}} \sqrt{\omega}}{j\omega L_E + \frac{K_E}{\sqrt{2}} \sqrt{\omega} + j \frac{K_E}{\sqrt{2}} \sqrt{\omega}} \quad (11)$$

Rearranging terms into real and imaginary (reactive) components, results in

$$Z_E(j\omega) = R_E + \frac{L_E^2 \frac{K_E}{\sqrt{2}} \omega^{5/2}}{N_E} + j \left[\omega L_{EB} + \frac{K_E^2 L_E \omega^2 + L_E^2 \frac{K_E}{\sqrt{2}} \omega^{5/2}}{N_E} \right] \quad (12)$$

where

$$N_E = K_E^2 \omega + \sqrt{2} K_E L_E \omega^{3/2} + L_E^2 \omega^2 \quad (13)$$

2.5. Modeling the Motional Impedance

The electrical motional impedance Z_M corresponding to Fig. 2, consists of three components in parallel: The capacitor $M_{MS}/(Bl)^2$, the resistor $(Bl)^2/R_{MS}$ and the inductor $(Bl)^2 C_{MS}$

$$Z_M = \frac{1}{j\omega C_p} \parallel R_p \parallel j\omega L_p \quad (14)$$

This can be converted to the familiar form

$$Z_M = \frac{R_p}{Q_{MS}} \cdot \frac{-j}{\frac{\omega}{\omega_0} - j \cdot \frac{1}{Q_{MS}} - \frac{\omega_0}{\omega}} \quad (15)$$

where:

$$Q_{MS} = C_p \cdot R_p \cdot \omega_0 \quad (16)$$

Q_{MS} is the mechanical Q-factor for the speaker. This can take the form

$$Z_M = R_M + jX_M \quad (17)$$

or

$$Z_M = |Z_M| \cos \theta + j |Z_M| \sin \theta \quad (18)$$

where

$$|Z_M| = \frac{Z_0 - R_E}{Q_{MS}} \cdot \frac{1}{\sqrt{\left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega}\right)^2 + \frac{1}{Q_{MS}^2}}} \quad (19)$$

and

$$\varphi = -\arctan\left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega}\right) Q_{MS} \quad (20)$$

From the aforementioned, the total impedance then becomes:

$$Z_S(f) = R_E + R_M + j(X_E + X_M) \quad (21)$$

and we find

$$|Z_S| = \sqrt{(R_E + R_M)^2 + (X_E + X_M)^2} \quad (22)$$

and

$$\theta = \arctan\left(\frac{X_E + X_M}{R_E + R_M}\right) \quad (23)$$

3. PRACTICAL OBSERVATIONS USING THE NEW MODEL

3.1. Example Using a 12 inch Woofer

Next, these formulas were used to model the impedance of an actual speaker. For this purpose a 12 inch Peerless XXLS subwoofer with aluminium cone was chosen, code 308 SWR 51 147 ALU 4L ALP 4 Ω . This is a driver with a heavy voice coil with high induction, so we should expect a ‘worst case’ discrepancy between traditional box simulation and actual box response.

Measurements were performed on this driver using the Brüel & Kjær Type 2012 Audio Analyzer and then transferred to the computer. Measured and curve fit data are both at the ISO R40 (1/12 octave) preferred frequencies. Results are presented on a logarithmic frequency axis and the ordinate axis the magnitude in dB scale. In this case the parameters were first measured in the traditional way and K_E , L_E and L_{EB} were found by iteration to obtain a best match between the measured and calculated curves. The parameters found by “traditional” method had finally to be fine adjusted by iteration to optimize the match. To this end, it was also necessary to introduce a R_E significantly higher than the measured R_{DC} . However, this yielded a nearly perfect match (see Fig. 5). Later, it will be shown how automatic curve fitting can be used to determine all the electrical parameters and establish a very close match, both in amplitude and phase.

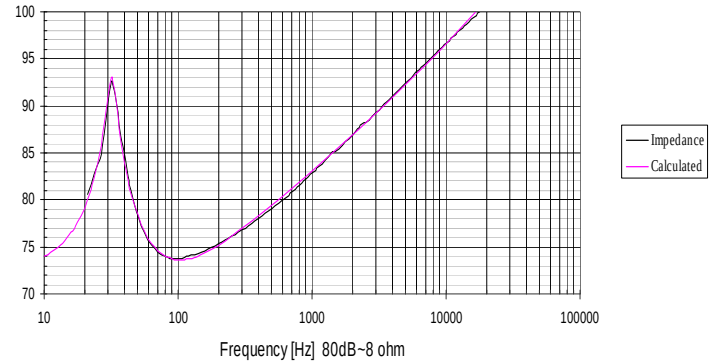


Fig. 5 Measured and modelled impedance data for the example loudspeaker driver. Ordinate is in dB, where $8\Omega = 80$ dB.

In this case, it was necessary to add 0.46Ω to R_{DC} to get the best fit. As R_{DC} was 2.8Ω , this represented an

increase of more than 16%. If this extra resistance is effective down to resonance (in this case, 32 Hz), it will have a significant impact on the electrical damping, and consequently also impact box simulation. To evaluate if this is the case, the blocked impedance was measured in the frequency range around resonance. The resistance was found to be 3.26Ω and not 2.8Ω . Therefore, we conclude that it is more accurate to calculate the Thiele Small parameters based upon $R_E = 2.8 + 0.46 = 3.26 \Omega$. This also generally confirms the need to distinguish between R_E and R_{DC} .

4. DEVELOPING A NEW IMPEDANCE MODEL

4.1. Applying Transformer Theory

The reason for this extra resistance is eddy currents. This can be shown by applying transformer circuit theory [5] explaining the discrepancy between the measured and predicted values of the impedance in the earlier example. Mutual coupling between the voice coil and other components in the loudspeaker driver has previously been observed [6, 7, 8, 9].

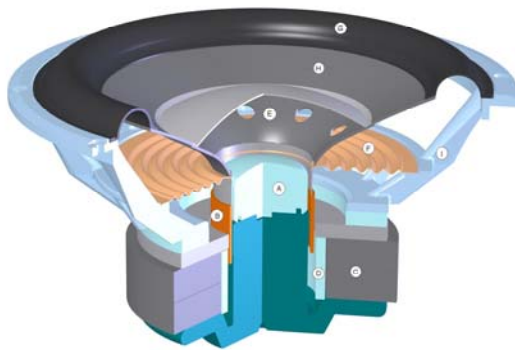


Fig.6 Cross-section of a subwoofer. Effectively, a transformer is created with the voice coil (B) as the primary and all of the conducting material around the voice coil as a one-turn secondary. Note the aluminium short-circuiting ring (D) and pole extension (A).

Fig. 6 shows a cross-section through a subwoofer similar to the one used in the measurements. This driver has an aluminium short-circuiting ring in the magnet system and an aluminium spacer on the pole piece to maximize symmetry.

If the iron core in the speaker were non-conductive, the result would be a resistor R_{DC} and an inductor L_0 , considerably larger than the L_E normally found for a loudspeaker (this has been proposed utilized to save the cost of the inductor in a cross-over network). Due to the conductivity of the core, we instead get a “transformer” with the n turns of the voice coil as primary and a rather special “one turn secondary”.

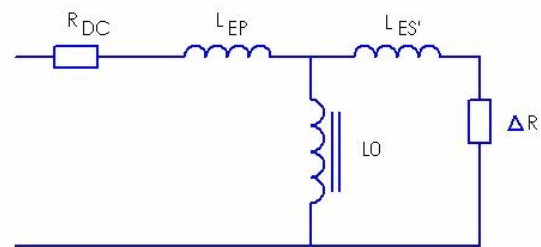


Fig.7 Equivalent circuit diagram for a transformer with turns ratio $n:1$ converted to primary side. $\Delta R = n^2 R_S$

Fig. 7 shows the equivalent circuit for a transformer with a turns ratio of $n:1$ shown with the secondary short-circuited, secondary winding resistance R_S and converted to primary side as ΔR . L_{EP} and $L_{ES'}$ are primary and secondary leakage inductances (the secondary converted to primary side). $L_{ES'}$ is L_{ES} converted to the primary side.

The same equivalent circuit is applicable for the blocked “speaker transformer” at low frequencies, where the skin depth is equal to the pole radius. The resistance of this secondary converted to the primary side is now ΔR and the secondary leak inductance converted to the primary side is L_{ES} . For this ‘transformer’, the lower cut off frequency occurs where the impedance ωL_0 equals ΔR . Below this frequency, the impact of ΔR becomes negligible as the impedance goes asymptotic toward R_{DC} . Above this very low frequency, the impact of L_0 , is negligible, so the total effective resistance R_E actually becomes the sum of ΔR and R_{DC} . This higher value has significant impact when used to calculate the Thiele-Small parameters. Above the cut off, frequency, the

primary and secondary leakage inductances are simply additive.

The primary leakage inductance in a transformer represents the part of the flux from the current in the primary winding not enclosed by the secondary – and vice versa. The tighter the coupling between the windings, the less leakage inductance is observed. So, thicker insulation between the windings results in more leakage inductance. The transformer analogy shows us that the traditional loudspeaker L_E is in fact the sum of the two ‘leakage inductances’.

Part of the primary leakage in the “speaker transformer” is due to the “overhung” part of the voice coil. Not all flux from the current through this section finds its way through the pole piece. Another cause of leakage is the spacing between the coil and the conductive pole piece, which acts like the insulation between the windings in a transformer. The inductor L_{EB} represents these components in the proposed equivalent circuit.

There is an additional and more significant source of leakage inductance. At low frequencies, the “skin depth” is equal to the radius of the pole piece. Eddy currents are distributed over the cross section of the pole piece. The current in the outermost part of the pole piece will enclose most of the AC flux through the pole piece. However, as the depth increases into the cross section of the pole piece, the current will enclose less and less of the flux from the voice coil current. A rigorous analysis of the distribution of flux and eddy currents can be examined by Finite Element Analysis. However, it is sufficient here to point out that not all of the primary flux through the pole piece is enclosed by all of the secondary current (eddy currents in the pole piece) and that this is the most significant source of leakage inductance. L_E' is caused by primary flux not enclosed by the secondary current and is best represented as an additional primary leakage inductance. Remember, however, that except at very low frequencies, contributions from the primary and secondary side are simply additive. As long as the “skin depth” is greater than $r/\sqrt{2}$ (where r is the radius of the pole piece), the distribution of flux and current is nearly constant, and consequently the resultant resistance and leakage inductance are also nearly constant. But as the skin depth diminishes, ΔR and $j\omega L_E'$ both increase as a function of ω . At the same time, the leakage inductance $L_E'(\omega)$ decreases with ω as the current in the progressively thinner conductive layer encloses an increasingly greater part of the AC flux

through the pole piece. This explains in principle the behaviour as predicted by Vanderkooy [4]. Around the transition frequency, $\Delta R + j\omega L_E'$ gradually changes from constant values for ΔR and L_{ES} to $K_E \cdot \sqrt{j\omega}$.

For model simplicity, ΔR is added to R_E , although clearly this is not absolutely correct. Keep in mind, however, that the goal here is to have a model that is both accurate and ‘practical’, i.e., not significantly more complex than absolutely necessary. Also note that this approximation has no significant negative consequences, as ΔR rapidly becomes a less significant part of K_E with increasing frequency. For practical use, L_0 can usually be neglected, so the circuit shown Fig. 4 can be used with $R_E = R_{DC} + \Delta R$, where R_E is determined by iteration or curve fitting. The exact fit is critical in the frequency range up to f_{min} .

To find the transition frequency for the example speaker between ΔR (0.46 ohm) and L_0 , we seek the frequency where

$$\omega L_0 = \Delta R \quad (23)$$

At this frequency the blocked impedance is close to being purely resistive

$$R_E = R_{DC} + \frac{\Delta R}{2} \quad (24)$$

In this example, the frequency was found to be at approximately 10 Hz, corresponding to an L_0 of 7.5 mH. The relative differential permeability is then about 3 and the transition frequency for semi-inductor behaviour should be expected to be around 27 Hz. The best method for deriving L_0 is using a curve fit, which will be described later.

4.2. Complete Practical Impedance Model

The total equivalent circuit diagram for the improved impedance model is shown Fig. 8. [10]

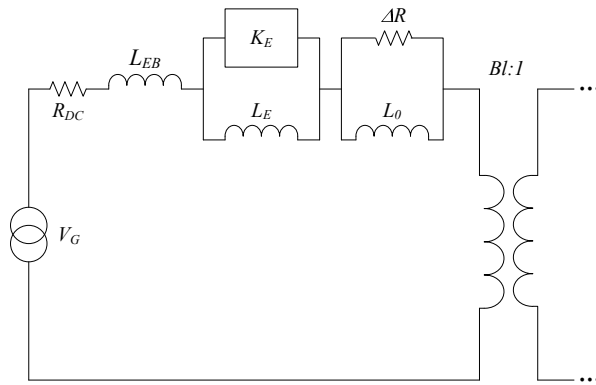


Fig. 8 New electrical equivalent circuit model.

The elements in this equivalent circuit are:

V_G	applied open circuit voltage
R_{DC}	DC resistance of the voice coil
Bl	motor force constant
L_{EB}	inductance of the part of the voice coil outside the motor gap
K_E	“semi-inductance” term related to eddy current and skin depth behavior in the motor, with the impedance of this element represented mathematically by $K_E \sqrt{j\omega}$
L_E	inductance of the part of the voice coil located inside the motor gap
L_0	inductance representing the coupling of the coil to the motor
ΔR	eddy current losses in the motor

Together, L_0 and ΔR represent that part of the low frequency behavior of the motor, in which the voice coil couples to the motor as if the motor were a single-turn resistive coil, and in which the skin depth is large. The K_E term, on the other hand, represents the higher frequency behavior of the motor, in which the skin depth is small [10].

The total electrical impedance is

$$Z_{E,Total} = Z_{E,Blocked} + (Bl)^2 z_m \quad (25)$$

where Z_m is the mechanical mobility of the transducer, representing all of the circuit load to the right of the transformer shown in Fig. 8 [1, 10].

5. VERIFICATION OF THE NEW MODEL

5.1. Measurements and Curve Fitting

A PC-based data acquisition system (Listen, Inc. SoundCheck) was used to measure the blocked impedance of a subwoofer at 1/12-octave intervals from 10 Hz to 20 kHz. The data were fit to the model using the Solver function in Microsoft Excel. When desired, introducing a high strength adhesive between the voice coil former and the pole piece of the drivers eliminated voice coil motion. This eliminated the effects of the back electro-motive force allowing the blocked impedance to be examined without having to remove the motional component of the electrical impedance.

The subwoofer tested was a 10” Peerless XXLS model 830843. This woofer features an aluminium spacer on the pole piece and an aluminium short circuiting ring. Table 1 shows the parameters used to fit the data to the circuit shown in Fig. 4. Fig. 9 shows the measured impedance magnitude and the curve fit based on the parameters in Table 1. Fig. 10 is similar, but shows the phase. This curve fit was based upon data taken with a blocked voice coil.

Parameter	Value	Units
R_E	6.19	ohms
L_{EB}	0.315	mH
K_E	0.134	Semi-Henrys
L_E	2.29	mH

Table 1. Peerless 830843 Blocked impedance parameters. Fit to the circuit shown in Fig. 4.

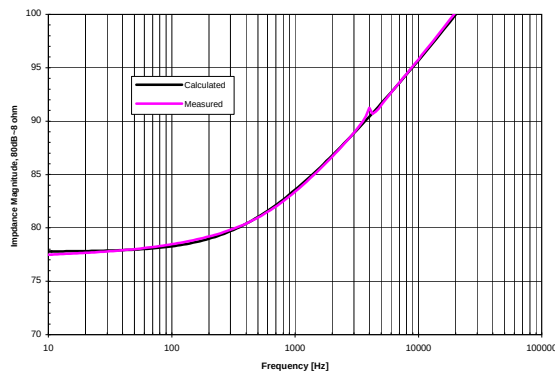


Fig. 9 Measured magnitude data and curve fit of the blocked impedance of the Peerless 830843.

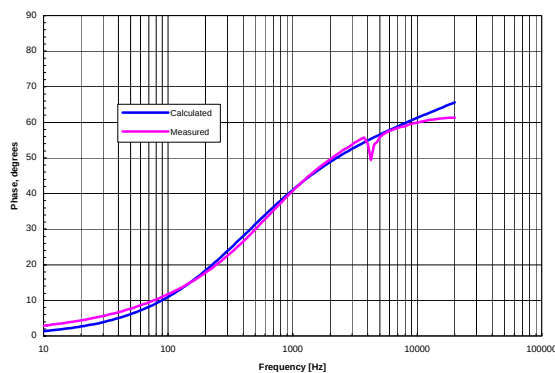


Fig. 10 Measured phase data and curve fit, blocked impedance – Peerless 830843.

As evidenced by Figs. 9 and 10, the curve fit is in excellent agreement with the measured data. The artifact found around 4 kHz in the measured data is the result of a mechanical resonance in the blocked system.

It is advantageous to be able to measure the blocked impedance parameters of a driver in a non-destructive fashion. To this end, the free air impedance of the driver was measured and curve fit to the ‘simplified’ blocked impedance model shown in Fig. 4. The motional component of the electrical impedance was fit using L_P , R_P , and C_P as discussed in Section 2.1. Table 2 shows the parameters used to fit the blocked impedance data.

Parameter	Value	Units
R_E	6.29	ohms
L_{EB}	0.319	mH
K_E	0.133	Semi-Henrys
L_E	2.38	mH
C_P	0.442	μ F
R_P	114	ohms
L_P	86.4	mH

Table 2. Peerless 830843 parameters. Blocked impedance fit to simplified circuit shown in Fig. 4. Unit was measured in free air.

Fig. 11 shows the measured impedance magnitude and curve fit based on the parameters in Table 2. Fig. 12 shows the phase.

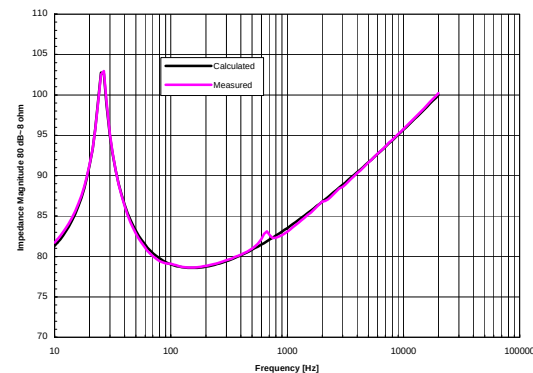


Fig. 11 Measured magnitude data and curve fit, Peerless 830843. Impedance fit to the simplified circuit shown in Fig. 4. Unit was measured in free air.

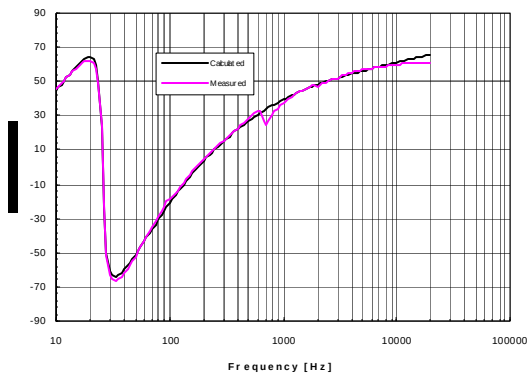


Fig. 12 Measured phase data and curve fit, Peerless 830843. Impedance fit to the simplified circuit shown in Fig. 4. Unit was measured in free air

Parameter	Value	Units
R_{DC}	5.83	ohms
L_{EB}	0.307	mH
K_E	0.138	Semi-Henrys
L_E	2.24	mH
L_0	10.8	mH
ΔR	0.540	ohms
C_P	0.445	μF
R_P	116	ohms
L_P	86.0	mH

Table 3. Peerless 830843, parameters. Impedance fit to the complete circuit shown in Fig. 8. Unit was measured in free air.

Again, the curve fit and the measured data are in excellent agreement showing that the voice coil does not need to be held motionless to accurately measure the blocked impedance parameters. Note that the blocked impedance parameters shown in Tables 1 and 2 show good agreement. The artifact found around 600 Hz in the measured data shown in Figs 11 through 14 is the

result of a rim resonance in the speaker's rubber surround.

Next, the data obtained by measuring the 830843 in free air was used to fit the 'complete' blocked impedance model shown in Fig. 8. Table 3 shows the results of this curve fit. Fig. 13 shows the measured impedance magnitude and the curve fit based on the parameters in Table 3. Fig. 14 is similar, but shows the phase.

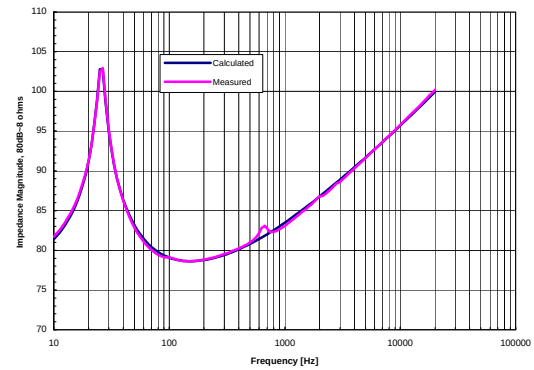


Fig. 13 Measured magnitude data and curve fit, Peerless 830843. Impedance fit to the complete circuit shown in Fig. 8. Unit was measured in free air.

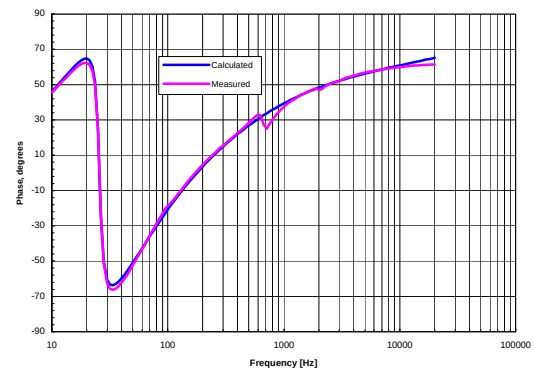


Fig. 14 Measured phase data and curve fit, Peerless 830843. Impedance fit to the complete circuit shown in Fig. 8. Unit was measured in free air

Compared to the data shown in Table 2, the data given in Table 3 produces a slightly better curve fit although the improvement is too small to be easily seen in the figures. The values for L_{EB} , K_E , and L_E are not strongly dependent upon which model is used – simplified or complete. In addition, these data show that it is not necessary to eliminate voice coil motion to measure the

blocked parameters as they can easily be derived from the free air impedance.

6. DISCUSSION

From a transformer point of view, the “speaker transformer” is very special. The pole piece is a rather poor quality iron core for a transformer, and in fact this can be a significant cause of nonlinear distortion, particularly as the core is pre-magnetized. In a real transformer, the core is either non-conductive or laminated to avoid eddy currents. However, in a loudspeaker, eddy currents actually provide a benefit by lowering the impedance rise due to inductance and extending the frequency range significantly at higher frequencies. This also provides extra power handling to the speaker, as some of the power is conducted to the “secondary winding”, heating the pole piece and not the voice coil [4]. Often, conducting material is introduced close to the coil in the form of aluminium (or copper) short circuiting rings inside the magnet system and/or a copper cap upon the pole piece to give better conductivity to the “single turn” secondary. The intended result is more output at higher frequencies, better power handling, and reduction of the AC magnetization of the pole piece, which in turn reduces distortion. By careful dimensioning it is possible not only to reduce the voice coil induction, but also to minimize its dependency on actual position. In this way, a primary cause of distortion in loudspeakers is reduced.

The impact of eddy currents in a copper cap, magnetic short-circuiting rings, and/or Cu/Al pole piece extensions will depend on how tightly they couple to the voice coil and the resistance they represent converted to the “primary” side. For a short-circuiting ring or pole piece extension, this is most often on the same order of magnitude as R_{DC} , however, the coupling is not very tight. With respect to the equivalent circuit, this means that $K_E \parallel L_E$ will be shunted with a rather large ‘leakage inductance’ in series with a resistor much larger than ΔR . As the voice coil still couples to the iron pole piece inside the air gap, the main effect is a slight reduction of Z_{min} and a reduction of K_E . Of course, the most important effect of the short-circuiting ring (with careful dimensioning) is reducing the L_E dependency on coil position – or at least making it more symmetric – and consequently reducing distortion.

A copper cap on the pole piece – and particularly in the extended “symmetric drive” version [15] – couples very tightly to the voice coil. However, as this takes up space

in the air gap, it is generally as thin as 0.3 mm, and therefore shunts $K_E \parallel L_E$ in the model with a very low leakage inductance in series with a large resistor. This effect can be observed as a rather flat impedance curve above a frequency where K_E is equal to this resistance. In general, loudspeakers having conducting rings in the magnet system still exhibit some semi-inductor behaviour in the impedance. However, K_E is reduced compared to same speaker without the conducting ring. The remaining semi-inductor effect must be due to the part of the coil inside the top plate, in close contact with magnetic conducting material both inside and around it (pole piece and top plate). It was found that the new equivalent circuit was also able to model speakers with conductive rings and copper caps.

The new equivalent circuit described here has already been successfully used to model the electrical impedance behaviour of the Tymphany Linear Array Transducer, and to determine the device’s Thiele-Small parameters [10]. In this case, the curve fitting was performed only on the amplitude, neglecting phase, as the functions are assumed to be “minimum phase”. This is the case as the equivalent circuit model is built up of simple passive components (e.g., resistors, inductors and capacitors). Even the semiconductor can be considered minimum phase, as it might be thought of as constructed of a series of shunt connections of inductors and resistors.

It should also be noted that the computer programme LspCAD, version 6.32, has been updated to handle the improved equivalent circuit model described here. Note that the model used in this programme includes an additional shunt resistor to L_{EB} not included in the equivalent circuit model described here. To force the programme to use the circuit model as exactly as described here, simply give this element a very high value to eliminate its effect. More information about this programme is available at: <http://www.jidata.com>. A simulation using this program is shown Fig. 12.

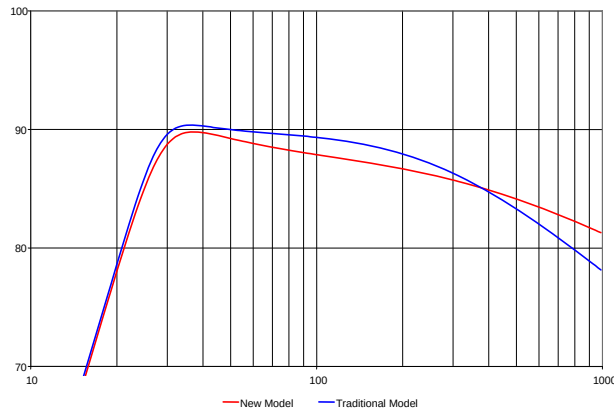


Fig. 12 Box simulation for the Peerless 830843 using the traditional and improved equivalent circuit models.

7. CONCLUSION

An improved blocked electrical impedance model for loudspeaker drivers was introduced that was found to be highly predictive of the actual behavior over a wide bandwidth - and therefore we conclude that the model is close to describing correctly the physical reality of the speakers blocked impedance. This model is applicable to most, if not all, electro dynamic transducers. It was shown that the voice coil actually behaves as the primary in a transformer, as it is inductively coupled to the surrounding conducting material, which in turn behaves as a one-turn secondary. Transformer theory was used to show how resistance and leakage inductance on the secondary side appears on the primary side, altering the expected values in the simple model. An improved model was proposed that introduced several new elements to account for the observed transformer behavior. For simplicity and ease of use, the transformer is removed from the model and elements on the secondary side are translated to their equivalent values on the primary side. The improved electrical equivalent circuit model described here could easily be incorporated into a complete equivalent circuit model using the electrical, mechanical and acoustic mobility (or impedance) analogies. A further advantage of the equivalent circuit proposed here, is that it also copes

well with loudspeakers having a very flat impedance curves.

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